

# Jian Wei CHEONG

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## Curriculum Vitae

### Professional Qualifications

- 2019 - 2023 **PhD, Physics**, *Nanyang Technological University*, Singapore  
2015 - 2019 **BSc, Physics, Honours**, *Nanyang Technological University*, Singapore  
2010 - 2013 **Diploma, Electrical Engineering**, *Ngee Ann Polytechnic*, Singapore

### Professional Experience

- 2024 - current **Research Fellow**, *Nanyang Technological University*, Singapore  
2023 **Project Officer**, *Nanyang Technological University*, Singapore  
2012 **Intern**, *ST Electronics*, Singapore  
**Teaching Assistant**, *Nanyang Technological University*, Singapore

| Year | Course                                               |
|------|------------------------------------------------------|
| 2026 | Physics Discovery Camp - Computational Nonlinear Lab |
| 2025 | Physics Discovery Camp - Computational Nonlinear Lab |
| 2025 | PH3101 Quantum Mechanics 2 (Restricted Repeat)       |
| 2024 | Physics Discovery Camp - Computational Nonlinear Lab |
| 2021 | PH3101 Quantum Mechanics 2                           |
| 2020 | PH1199 Physics Lab 1B                                |
| 2019 | PH1198 Physics Lab 1A                                |

### Awards & Achievements

- 2019 **Short-speech Contest Best Presentation** (PAP701 Graduate seminar module), *Nanyang Technological University*, Singapore  
2017/2018 **Dean's List** (top 5% of cohort), *Nanyang Technological University*, Singapore  
2016/2017 **NTU President Research Scholar** (completing URECA), *Nanyang Technological University*, Singapore  
2011 **Director's List** (top 5% of cohort), *Ngee Ann Polytechnic*, Singapore  
2011 **Best Performance, Programmable Logic Device** (top student of cohort), *Ngee Ann Polytechnic*, Singapore  
2010 **Best Performance, Digital Electronics & Practice** (top student of cohort), *Ngee Ann Polytechnic*, Singapore

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## Publications

1. Y. Wang, A. Pradana, [J. W. Cheong](#), and L. Y. Chew, **Enhancing performance and operational regimes of the quantum Otto refrigerator with heat bath algorithmic cooling**, *Physical Review E*, **113**(5), 054112 (2026).
2. [J. W. Cheong](#), A. Pradana, and L. Y. Chew, **Enhancing the performance of quantum switch refrigeration with an information pool of qubits**, *Physical Review Research*, **8**(1), 013272 (2026).
3. [J. W. Cheong](#), A. Pradana, and L. Y. Chew, **Non-Markovian refrigeration and heat flow in the quantum switch**, *Physical Review A*, **110**(2), 022220 (2024).
4. L. Y. Chew, A. Pradana, L. He, and [J. W. Cheong](#), **Stochastic thermodynamics of finite-tape information ratchet**, *European Physical Journal Special Topics* (2023).
5. [J. W. Cheong](#), A. Pradana, and L. Y. Chew, **Effects of non-Markovianity on daemonic ergotropy in the quantum switch**, *Physical Review A*, **108**(1), 012201 (2023).
6. L. He, [J. W. Cheong](#), A. Pradana, and L. Y. Chew, **Effects of correlation in an information ratchet with finite tape**, *Physical Review E*, **107**(2), 024130 (2023).
7. [J. W. Cheong](#), A. Pradana, and L. Y. Chew, **Communication advantage of quantum compositions of channels from non-Markovianity**, *Physical Review A*, **106**(5), 052410 (2022).
8. L. He, A. Pradana, [J. W. Cheong](#), and L. Y. Chew, **Information processing second law for an information ratchet with finite tape**, *Physical Review E*, **105**(5), 054131 (2022).

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## Conferences

### Proceedings

1. L. Y. Chew, [J. W. Cheong](#), and A. Pradana, **Thermodynamic Functionality of Non-detailed Balance Finite-Tape Information Ratchet**, *International Conference on Geometric Science of Information*, Springer Nature Switzerland, pp. 173–181 (2025).

### Talks

1. [J. W. Cheong](#)<sup>†</sup>, A. Pradana, and L. Y. Chew, **Maxwell's Uncertain Demon: Achieving Nonclassical and Enhanced Refrigeration when Causality Blurs**, *Asia-Pacific Conference on Networks and Complex Systems 2026 (APCNCS 2026)*, Singapore, June 2026.

### Posters

1. [J. W. Cheong](#), A. Pradana, and L. Y. Chew<sup>†</sup>, **Enhancement of quantum processes from indefinite causal order through non-Markovianity**, *29th International Conference on Statistical Physics (STATPHYS29)*, Florence, Italy, July 2025.
2. [J. W. Cheong](#)<sup>†</sup>, A. Pradana, and L. Y. Chew, **Non-Markovian refrigeration and heat flow in the quantum switch**, *Quantum Thermodynamics Conference 2025 (QTD2025)*, Singapore, July 2025.
3. L. Y. Chew<sup>†</sup>, [J. W. Cheong](#), and A. Pradana, **Enhancement of quantum processes from indefinite causal order through non-Markovianity**, *XLV Dynamics Days Europe 2025 (DDE2025)*, Thessaloniki, Greece, June 2025.

4. L. Y. Chew<sup>†</sup>, L. He, A. Pradana, and J. W. Cheong, **Stochastic thermodynamics and correlation effects of finite-tape information ratchets**, *28th International Conference on Statistical Physics (STATPHYS28), Tokyo, Japan, August 2023*.

<sup>†</sup>Presenting author.

## Technical Experience

|             | Skill                       | Level                    | Comment                                               |
|-------------|-----------------------------|--------------------------|-------------------------------------------------------|
| Programming | Python                      | ■■■■                     | 9+ years experience, used in main work                |
|             | Julia                       | ■■■■                     | 4+ years experience, used in main work                |
|             | R                           | ■■■□                     | graduate course, computational biology, data analysis |
|             | C / C++                     | ■■■□                     | bachelor course                                       |
|             | MATLAB                      | ■■■□                     | bachelor course                                       |
|             | Haskell                     | ■□□□                     | personal quantum computing projects                   |
|             | Racket                      | ■□□□                     | personal quantum computing projects                   |
|             | Common Lisp                 | ■□□□                     | personal software projects                            |
|             | Bash / sh                   | ■■■□                     | personal Linux projects                               |
|             | Quarto                      | ■■■□                     | website, presentations, and reports                   |
|             | LaTeX / Typst               | ■■■□                     | presentations, reports, and published papers          |
|             | HTML / CSS                  | ■■■□                     | personal website                                      |
|             | ■□□□ basic knowledge        | ■■■□ extensive knowledge |                                                       |
|             | ■■■□ intermediate knowledge | ■■■■ expert knowledge    |                                                       |

## Miscellaneous Projects

- **Strain estimation for hazard forecastings before and after 2011 Japan Tohoku earthquake** by analysis of seismic GPS displacement data and computation of seismic strains from velocity fields.  
*ES7008 Geophysical Data Analysis, NTU*
- **Variations in statistical complexity of genome sequences across species** by analysis of genome sequences from GenBank assembly with Baum-Welch algorithm.  
*CE7412 Computational and Systems Biology, NTU*
- **Detecting adversarial attack of deep neural networks for image recognition from image complexity** by Fast Gradient Sign Method (FGSM), DeepFool, One Pixel Attack, and Jacobian-Based Saliency Map Attack (JSMA).  
*PH3502 Chaotic Dynamical Systems, NTU*
- **Monte Carlo photon transport in multi-layered biological tissues** with Henyey-Greenstein scattering for diagnostic imaging and photon therapy.  
*PH4505 Computational Physics, NTU*
- **Monte Carlo simulation of periodic-driven Brownian particles** with Arrhenius equation, demonstrating dissipation-driven adaptation.  
*PAP723 Numerical Methods for Physicists, NTU*

- **Designing, programming, 3D printing, and building a hovering quadcopter drone** with Arduino, Autodesk EAGLE, Fusion360, and COMSOL Multiphysics.  
*Making and Tinkering Lite 1, NTU*